Equivalent Fractions

Cut every piece the same way: more pieces, same amount.

2/3: cut every piece into 4



$$\frac{2}{3} = \frac{8}{12}$$

4 times as many pieces, and 4 times as many shaded.

Multiply the top and the bottom by the same number

top: $2 \times 4 = 8$

bottom: $3 \times 4 = 12$

$$\frac{2}{3} = \frac{8}{12}$$

Divide to get bigger pieces



$$\frac{6}{10} = \frac{3}{5}$$

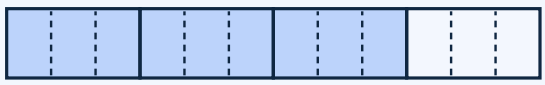
Put the pieces in pairs: top ÷ 2 and bottom ÷ 2.

Hundredths: tenths cut into 10

$$\frac{3}{10} = \frac{30}{100}$$

Every tenth is 10 hundredths.

Your turn!



$$\frac{3}{4} = \frac{\square}{\square}$$

Equivalent Fractions: Practice

Do the same to the top and the bottom.

Part 1 Cut the pieces Count the small pieces, then the shaded ones.

1) Each piece is cut into 4.



$\frac{1}{3} = \frac{\square}{\square}$

2) Each piece is cut into 2.



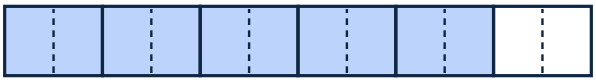
$\frac{3}{5} = \frac{\square}{\square}$

3) Each piece is cut into 3.



$\frac{2}{4} = \frac{\square}{\square}$

4) Each piece is cut into 2.



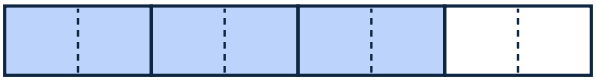
$\frac{5}{6} = \frac{\square}{\square}$

5) Each piece is cut into 6.



$\frac{1}{2} = \frac{\square}{\square}$

6) Each piece is cut into 2.



$\frac{3}{4} = \frac{\square}{\square}$

Part 2 Times the top and the bottom Multiply both by the number shown.

7) times 2: $\frac{2}{5} = \frac{\square}{\square}$

8) times 3: $\frac{1}{4} = \frac{\square}{\square}$

Equivalent Fractions: Practice (continued)

Part 2 Times the top and the bottom (continued)

9) times 2: $\frac{5}{6} = \frac{\square}{\square}$

10) times 4: $\frac{2}{3} = \frac{\square}{\square}$

11) times 4: $\frac{1}{3} = \frac{\square}{\square}$

12) times 10: $\frac{3}{10} = \frac{\square}{\square}$

13) times 6: $\frac{1}{2} = \frac{\square}{\square}$

14) times 20: $\frac{4}{5} = \frac{\square}{\square}$

15) times 2: $\frac{1}{6} = \frac{\square}{\square}$

16) times 10: $\frac{9}{10} = \frac{\square}{\square}$

Part 3 Find the top number No help this time.

17) $\frac{3}{4} = \frac{\square}{12}$

18) $\frac{1}{3} = \frac{\square}{12}$

19) $\frac{5}{6} = \frac{\square}{12}$

Equivalent Fractions: Practice (continued)

Part 3 Find the top number (continued)

20) $\frac{1}{4} = \frac{\square}{12}$

21) $\frac{7}{10} = \frac{\square}{100}$

22) $\frac{1}{2} = \frac{\square}{100}$

23) $\frac{2}{3} = \frac{\square}{12}$

24) $\frac{3}{5} = \frac{\square}{100}$

25) $\frac{1}{6} = \frac{\square}{12}$

26) $\frac{1}{4} = \frac{\square}{100}$

27) $\frac{1}{2} = \frac{\square}{12}$

28) $\frac{2}{5} = \frac{\square}{100}$

Part 4 Find the bottom number No help this time.

29) $\frac{3}{4} = \frac{75}{\square}$

30) $\frac{2}{3} = \frac{8}{\square}$

31) $\frac{1}{5} = \frac{20}{\square}$

Equivalent Fractions: Practice (continued)

Part 4 Find the bottom number (continued)

32) $\frac{5}{6} = \frac{10}{\square}$

33) $\frac{3}{10} = \frac{30}{\square}$

34) $\frac{1}{2} = \frac{6}{\square}$

35) $\frac{3}{4} = \frac{9}{\square}$

36) $\frac{4}{5} = \frac{80}{\square}$

37) $\frac{9}{10} = \frac{90}{\square}$

Part 5 Fewer, bigger pieces No help this time. Divide the top and the bottom by the same number.

38) $\frac{8}{12} = \frac{\square}{3}$

39) $\frac{60}{100} = \frac{\square}{5}$

40) $\frac{9}{12} = \frac{\square}{4}$

41) $\frac{50}{100} = \frac{\square}{2}$

42) $\frac{10}{12} = \frac{\square}{6}$

43) $\frac{40}{100} = \frac{\square}{5}$

Equivalent Fractions: Practice (continued)

Part 5 Fewer, bigger pieces (continued)

44) $\frac{4}{12} = \frac{\square}{3}$

45) $\frac{75}{100} = \frac{\square}{4}$

46) $\frac{6}{12} = \frac{\square}{2}$

47) $\frac{20}{100} = \frac{\square}{5}$

48) $\frac{2}{12} = \frac{\square}{6}$

49) $\frac{70}{100} = \frac{\square}{10}$

Part 6 Equal or not? Circle yes or no.

50) $\frac{2}{3}$ and $\frac{8}{12}$ yes no

51) $\frac{3}{4}$ and $\frac{6}{12}$ yes no

52) $\frac{1}{5}$ and $\frac{20}{100}$ yes no

53) $\frac{5}{6}$ and $\frac{10}{12}$ yes no

54) $\frac{2}{5}$ and $\frac{25}{100}$ yes no

55) $\frac{3}{10}$ and $\frac{30}{100}$ yes no

Equivalent Fractions: Practice (continued)

Part 6 Equal or not? (continued)

56) $\frac{1}{4}$ and $\frac{4}{12}$ yes no

57) $\frac{7}{10}$ and $\frac{70}{100}$ yes no

Part 7 Equal fractions

58) $\frac{3}{5} = \frac{\square}{15}$

59) $\frac{1}{4} = \frac{\square}{16}$

60) $\frac{2}{4} = \frac{\square}{20}$

61) $\frac{3}{4} = \frac{\square}{16}$

62) $\frac{4}{5} = \frac{\square}{15}$

63) $\frac{3}{5} = \frac{\square}{25}$

64) $\frac{2}{5} = \frac{\square}{25}$

65) $\frac{1}{6} = \frac{\square}{30}$

66) $\frac{3}{6} = \frac{\square}{30}$

Equivalent Fractions: Practice (continued)

Part 7 Equal fractions (continued)

67) $\frac{5}{6} = \frac{\square}{24}$

68) $\frac{1}{5} = \frac{\square}{15}$

69) $\frac{2}{6} = \frac{\square}{18}$

70) $\frac{1}{6} = \frac{\square}{18}$

71) $\frac{4}{5} = \frac{\square}{20}$

72) $\frac{1}{3} = \frac{\square}{15}$

73) $\frac{2}{5} = \frac{\square}{15}$

74) $\frac{1}{5} = \frac{\square}{25}$

75) $\frac{2}{4} = \frac{\square}{16}$

76) $\frac{4}{6} = \frac{\square}{24}$

77) $\frac{2}{6} = \frac{\square}{24}$

78) $\frac{1}{6} = \frac{\square}{24}$

Equivalent Fractions: Practice (continued)

Part 7 Equal fractions (continued)

79) $\frac{2}{6} = \frac{\square}{30}$

80) $\frac{3}{5} = \frac{\square}{20}$

81) $\frac{3}{6} = \frac{\square}{18}$

Part 8 Make it simpler

82) $\frac{20}{24} = \frac{\square}{6}$

83) $\frac{12}{16} = \frac{\square}{4}$

84) $\frac{4}{24} = \frac{\square}{6}$

85) $\frac{12}{20} = \frac{\square}{5}$

86) $\frac{3}{18} = \frac{\square}{6}$

87) $\frac{9}{15} = \frac{\square}{5}$

88) $\frac{15}{18} = \frac{\square}{6}$

89) $\frac{3}{15} = \frac{\square}{5}$

90) $\frac{4}{20} = \frac{\square}{5}$

Equivalent Fractions: Practice (continued)

Part 8 Make it simpler (continued)

91) $\frac{8}{20} = \frac{\square}{5}$

92) $\frac{12}{15} = \frac{\square}{5}$

93) $\frac{4}{16} = \frac{\square}{4}$

94) $\frac{6}{15} = \frac{\square}{5}$

95) $\frac{16}{20} = \frac{\square}{5}$

Part 9 Remember? Problems from earlier packets.

96) $1{,}248 \div 4 = \square$

97) $42 \times 13 = \square$

98) $784 \div 4 = \square$

99) $26 \times 3 = \square$

100) 5,912 to the nearest hundred = $\square$

101) $\frac{4}{5}$ of 25 = $\square$

Equivalent Fractions: Practice (continued)

Part 10 Word problems Draw a strip and cut the pieces.

102) Mia ate 3/4 of a pizza cut in 12 slices. How many slices did she eat?

Number sentence _____

Answer ☐ slices

Draw or show your work

103) A fence has 100 boards. 3/10 of them are painted. How many boards are painted?

Number sentence _____

Answer ☐ boards

Draw or show your work

104) Leo read 8/12 of his book. Kim read 2/3 of the same book. Did they read the same amount?

Number sentence _____

Answer ☐

Draw or show your work

105) 60 of 100 seats are full. What fraction is that, in fifths?

Number sentence _____

Answer ☐ full

Draw or show your work

106) Challenge: Sam says 2/5 = 4/10 = 40/100. Is he right? What would 2/5 be in twentieths?

Number sentence _____

Answer ☐

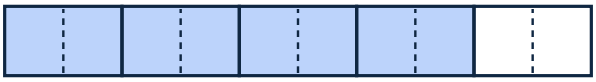
Draw or show your work

Check-In: Equivalent Fractions

Show what you know. No help this time. Score: _____ / 9

Part 1 Cut the pieces

1) Each piece is cut into 2.



$\frac{4}{5} = \frac{\square}{\square}$

Part 2 Find the top number

2) $\frac{4}{6} = \frac{\square}{12}$

3) $\frac{3}{6} = \frac{\square}{12}$

4) $\frac{4}{5} = \frac{\square}{100}$

Part 3 Fewer, bigger pieces

5) $\frac{50}{100} = \frac{\square}{4}$

6) $\frac{60}{100} = \frac{\square}{10}$

7) $\frac{6}{12} = \frac{\square}{4}$

Part 4 Equal or not? Circle yes or no.

8) $\frac{4}{6}$ and $\frac{8}{12}$ yes no

9) $\frac{2}{4}$ and $\frac{5}{8}$ yes no